

FREDRICK MANDELA

ENGINEERING LEAD | AI AUTOMATION & DATA PLATFORMS

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PROFESSIONAL SUMMARY

Engineering lead with 6+ years of experience building production automation, data platforms, integrations, APIs, and internal tooling, including leading engineers supporting multi-country operations. Strong hands-on background in Python, SQL, workflow orchestration, data quality, APIs, and production reliability, with recent work building AI-assisted automation using LLMs, structured outputs, validation, document extraction, and multi-channel integrations. Experienced translating operational problems into reliable technical systems, prioritizing competing stakeholder needs, and balancing rapid experimentation with production controls.

CORE COMPETENCIES

AI Automation & LLM Workflows | Python & API Integrations | Workflow Orchestration | Engineering Leadership | Production Reliability & Observability | Internal Platforms & Automation | Structured AI Outputs & Validation | Data Platforms & PostgreSQL | CI/CD & Cloud Deployment

PROFESSIONAL EXPERIENCE

ONE ACRE FUND, Nairobi, Kenya

Data Engineering Manager / Lead Data Engineer | Jan 2021 - Mar 2026

- Led a team of 3 data engineers and a business analyst delivering production data platforms, automations, integrations, and internal data products across 10+ country operations.
- Owned engineering prioritization and delivery across competing product, operations, finance, and analytics requirements; reviewed technical designs, removed blockers, and coordinated production releases.
- Designed API- and database-driven automations connecting operational systems, payment platforms, field applications, and analytics environments, reducing manual data movement and improving data availability.
- Built and operated data infrastructure supporting 500M+ records and 5TB+ across Snowflake, SQL Server, Dataiku, Airflow, dbt, and Superset.
- Established monitoring, automated data-quality checks, incident follow-up, and clearer ownership practices to improve reliability and operational visibility across production workflows.
- Supported large-scale migration and synchronization programs and developed reusable processes for onboarding new countries while reducing manual cleanup.
- Implemented governance and access patterns including role-based controls, row-level security, validation, and documentation for business users.

ONE ACRE FUND, Nairobi, Kenya

Data Engineer - IT Engineering | Jan 2018 - Dec 2020

- Designed ETL/ELT pipelines ingesting data from APIs, relational databases, operational systems, and third-party platforms into centralized warehouse environments.
- Developed API- and database-driven integrations using Python, SQL, C#, and workflow automation tools to improve data availability across business systems.
- Helped transition legacy SSIS workflows to modern automated pipelines, improving scalability, maintainability, monitoring, and refresh reliability.
- Partnered with BI, product, and business stakeholders to define requirements, validate outputs, and deliver accurate datasets for operational and strategic decisions.

AI & AUTOMATION PROJECTS

INVESTAR COMPANION - AI-ASSISTED FINANCIAL DATA AUTOMATION | Python, OpenAI API, PostgreSQL/Supabase, Playwright, Pydantic, GitHub Actions

- Designed and built an AI-assisted automation platform that acquires Kenyan investment-fund data from websites, email, PDFs, spreadsheets, and structured files, normalizes it into a canonical data model, and distributes market updates.
- Built a hybrid ingestion architecture that routes stable sources through deterministic parsers and irregular or changing documents through structured AI extraction, balancing reliability, cost, and flexibility.
- Added guardrails including Pydantic-validated structured outputs, confidence scores, source evidence, SHA-256 content hashes, canonical entity resolution, and auditable extraction metadata.
- Automated acquisition, AI extraction, validation, persistence, analysis, AI-generated content, and publishing through modular adapters for Slack, email, WhatsApp, and Telegram; isolated channel failures to preserve workflow resilience.

AI SQL ANALYST - LLM DATA ASSISTANT | Python, OpenAI API, SQL, Snowflake

- Built an LLM workflow that translates natural-language questions into SQL using database schema context, validates generated queries before execution, runs them against structured data, and explains results for non-technical users.

TECHNICAL PROFICIENCIES

AI & Automation: OpenAI API, structured LLM outputs, AI-assisted development, document extraction, workflow automation, prompt design and evaluation

Programming & Integration: Python, SQL, C#, REST APIs, webhooks, Playwright

Orchestration & Data: Airflow, dbt, Dataiku, GitHub Actions, Kafka, RabbitMQ, Airbyte, Snowflake, PostgreSQL, Supabase, SQL Server, MongoDB, Elasticsearch

Reliability & Governance: Pydantic validation, Soda, pipeline monitoring, data quality, metadata management, RBAC, row-level security, Slack/email alerts

Cloud & DevOps: Azure, AWS, Docker, Git, GitHub, CI/CD practices

EARLIER EXPERIENCE

One Acre Fund - Finance Systems Data Integration Intern | Mar 2017 - Aug 2017 | Built ETL workflows and optimized warehouse structures for analytics.

PwC - Technology Advisory Intern, Boston, USA | 2015 - 2016 | Supported technology consulting, systems integration planning, KPI analysis, and data migration engagements.

EDUCATION

Bachelor of Science (B.Sc.) in Mathematics & Computer Science | Brown University | 2013 - 2017